



Lab 02

Introduction to Packet Tracer and IP Routing

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2.1 Objective

The Objectives of this lab are:

- Implementing static routing with a two-router configuration.
- Implementing static routing with a three-router configuration.
- Implementing dynamic routing using OSPF.

2.2 Background/Scenario

In computer networks, routing is the process of selecting paths for traffic in a network or between or across multiple networks. Understanding how to effectively manage and control this process is crucial for network administrators. There are two primary methods of routing: static routing and dynamic routing.

Static Routing is a method where the network administrator manually enters routes into the routing table. These routes do not change unless they are manually updated by the administrator. Static routing is straightforward and can be more secure since it doesn't rely on external protocols to manage routes. However, it lacks scalability and flexibility, making it suitable for small or simple networks where the network topology is unlikely to change frequently.

Dynamic Routing, on the other hand, uses routing protocols to automatically discover and update routes in the routing table. Protocols like OSPF (Open Shortest Path First) enable routers to communicate with each other, share information about network topology, and automatically adjust to changes, such as link failures or changes in the network structure. Dynamic routing is essential for larger, more complex networks where manual updates would be impractical.

In this lab, you will explore both static and dynamic routing, understanding how to configure them,



Topology

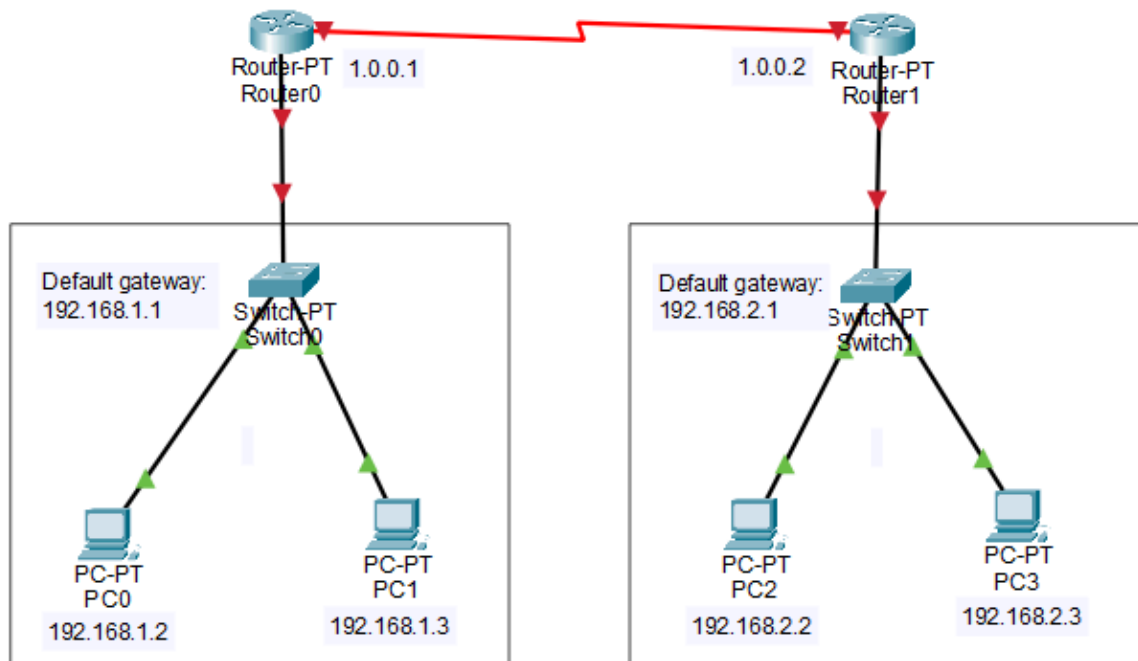


Figure 1

Note: When assigning addresses to interfaces, make sure your interface is the same as the one that needs to be assigned. Interfaces in your topology might have different numbering depending on which wire you place first.

Enable interfaces by pressing CTRL + R and check the 'Always show port labels in logical workspace' option.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	Se2/0	1.0.0.1	255.255.255.0	N/A	N/A
	FastEthernet0/0	192.168.1.1	255.255.255.0	N/A	S0 F0/1



R2	Serial2/0	1.0.0.2	255.255.255.0	N/A	N/A
	FastEthernet0/0	192.168.2.1	255.255.255.0	N/A	S1 F0/1
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1	S1 F1/1
PC-B	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F2/1
PC-C	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F1/1
PC-D	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S1 F2/1

Table 1: IP addressing table

Task 1: Implementing Static Routing in a Two-Router Configuration

1. Construct the topology shown above
2. Configure the PCs with the IP addresses and subnet masks according to the IP address table given above
 - a. To assign an IP address to a PC, click on the PC, go to Desktop and then IP configuration. Fill up the IP address and subnet mask.



PC-A

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:F3FF:FE2E:7B6D

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Figure 2

- b. Alternatively, you can also use the ipconfig command in the PC terminal
ipconfig <IPv4 address><subnet mask><default gateway>

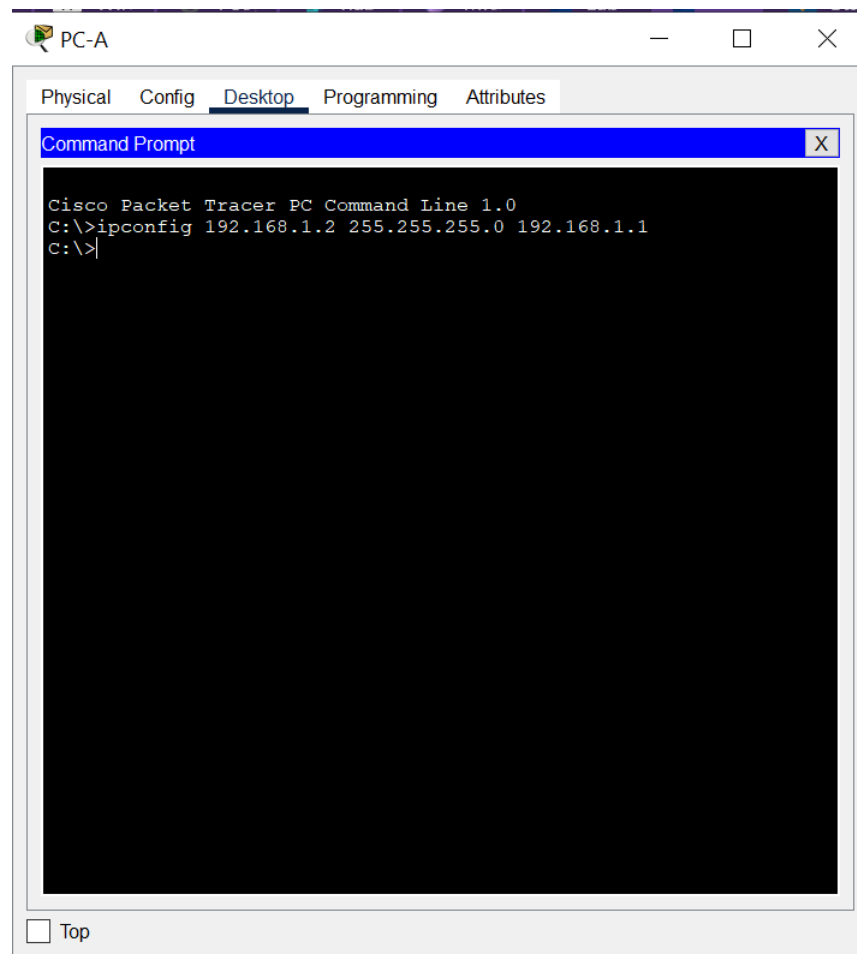


Figure 3

3. Configure routers with IP addresses and subnet masks
 - a. To assign an IP address to a router, go to the router's config tab and then interfaces. Configure the IP address in Fastethernet and Serial ports according to Table 1.



Router1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0060.3EB1.ECBA

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with
CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
```

☐ Top

Figure 4

4. Assign routes to the routers
 - a. Go to the router's CLI, then type the following commands
ip route <network id><subnet mask> <next hop>

Static route for Router1:

Router(config-if)#ip route 192.168.2.0 255.255.255.0 1.0.0.2

**Static route for Router2:**

Router(config-if)#ip route 192.168.1.0 255.255.255.0 1.0.0.1

Attach screenshots of the configurations.

Router 1

The screenshot shows the configuration window for Router0, specifically the 'Config' tab for the 'FastEthernet0/0' interface. The left sidebar shows a tree view with 'GLOBAL' and 'INTERFACE' sections. Under 'INTERFACE', 'FastEthernet0/0' is selected. The main area displays the configuration for this interface, including 'Port Status' (On), 'Link Speed' (100 Mbps), 'Duplex' (Full Duplex), 'MAC Address' (0009.7C38.1909), 'IP Configuration' (IPv4 Address: 192.168.1.1, Subnet Mask: 255.255.255.0), and 'Tx Ring Limit' (10). At the bottom, the 'Equivalent IOS Commands' section shows the following commands:

```
Router(config)#  
Router(config)#interface FastEthernet0/0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial2/0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#  
Router(config)#  
Router(config)#  
Router(config)#interface FastEthernet0/0  
Router(config-if)#
```



Router0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 1.0.0.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial2/0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#  
Router(config)#  
Router(config)#  
Router(config)#interface FastEthernet0/0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial2/0  
Router(config-if)#
```

Top



Router0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Static Routes

Network

Mask

Next Hop

Add

Network Address

192.168.2.0/24 via 1.0.0.2

Remove

Equivalent IOS Commands

```
Router (config-if)#  
Router (config-if)#exit  
Router (config)#  
Router (config)#  
Router (config)#  
Router (config)#interface FastEthernet0/0  
Router (config-if)#  
Router (config-if)#exit  
Router (config)#interface Serial2/0  
Router (config-if)#  
Router (config-if)#exit  
Router (config)#  
Router (config)#  
Router (config)#
```

Router 2



Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.8F17.EAB0

IP Configuration

IPv4 Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
```

Top



Router1

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 1.0.0.2

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
```



Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Static Routes

Network

Mask

Next Hop

Add

Network Address

192.168.1.0/24 via 1.0.0.1

Remove

Equivalent IOS Commands

```
Router1>show ip route. Line protocol on Interface Serial2/0, changed state to up
Router1>
Router (config-if) #
Router (config-if) #exit
Router (config) #interface FastEthernet0/0
Router (config-if) #
Router (config-if) #
Router (config-if) #exit
Router (config) #interface Serial2/0
Router (config-if) #
Router (config-if) #exit
Router (config) #
Router (config) #
Router (config) #
```

5. Verify the network by pinging the IP address of any PC.
 - a. Click on PC, then go to the command prompt
 - b. Type 'ping <ip address of targeted node>'



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=2ms TTL=126
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms

C:\>
```

Figure 5



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=5ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>
```



Topology

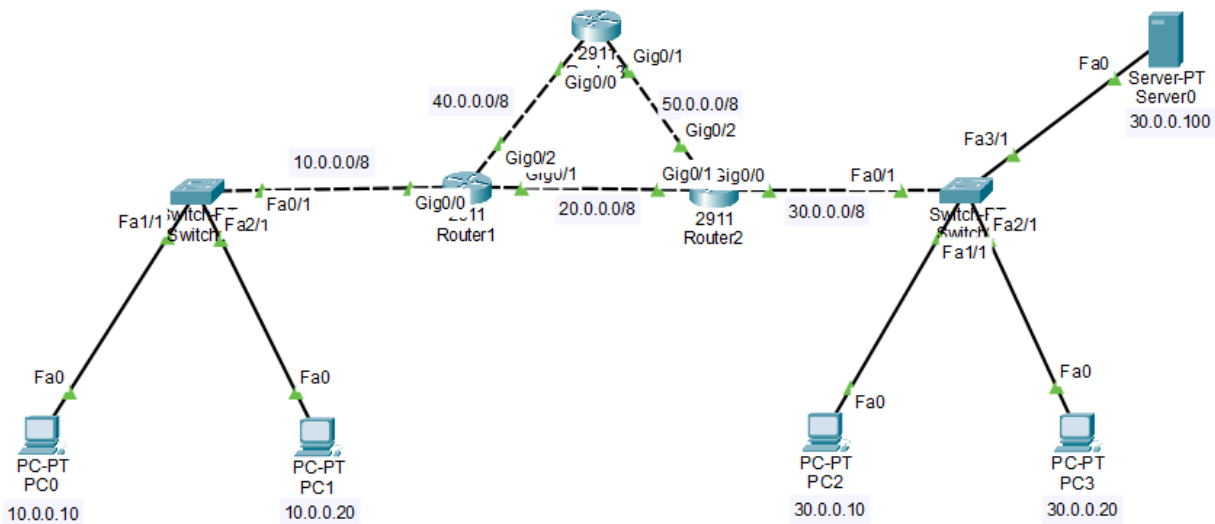


Figure 6

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/0	10.0.0.1	255.0.0.0	N/A	S0 F0/1
	G0/1	20.0.0.1	255.0.0.0	N/A	N/A
	G0/2	40.0.0.1	255.0.0.0	N/A	N/A
R2	G0/0	30.0.0.1	255.0.0.0	N/A	S1 F0/1
	G0/1	20.0.0.2	255.0.0.0	N/A	N/A
	G0/2	50.0.0.2	255.0.0.0	N/A	N/A
R3	G0/0	40.0.0.2	255.0.0.0	N/A	N/A



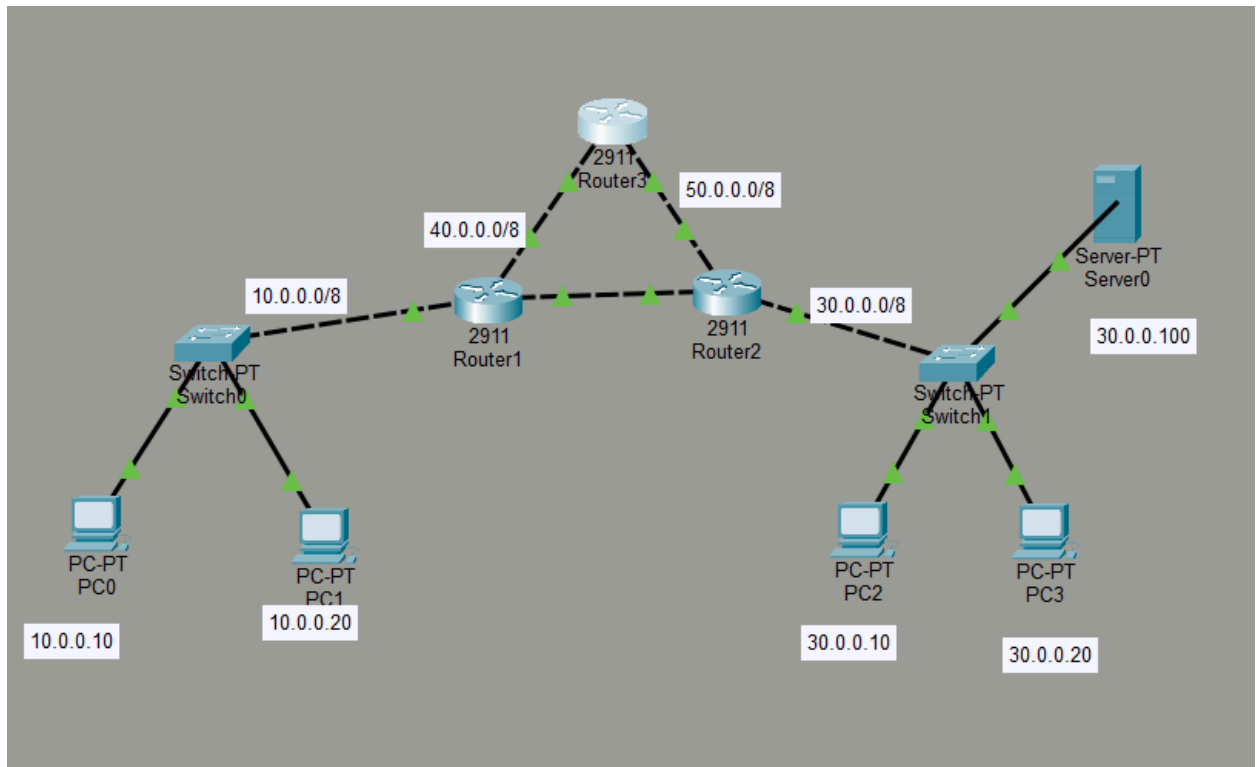
	G0/1	50.0.0.1	255.0.0.0	N/A	N/A
PC-1	NIC	10.0.0.10	255.0.0.0	10.0.0.1	S1 F1/1
PC-2	NIC	10.0.0.20	255.0.0.0	10.0.0.1	S1 F2/1
PC-3	NIC	30.0.0.10	255.0.0.0	30.0.0.1	S2 F1/1
PC-4	NIC	30.0.0.20	255.0.0.0	30.0.0.1	S2 F2/1
Server-S0	NIC	30.0.0.100	255.255.255.0	30.0.0.1	S2 F3/1

Task 2 – IP Addressing table



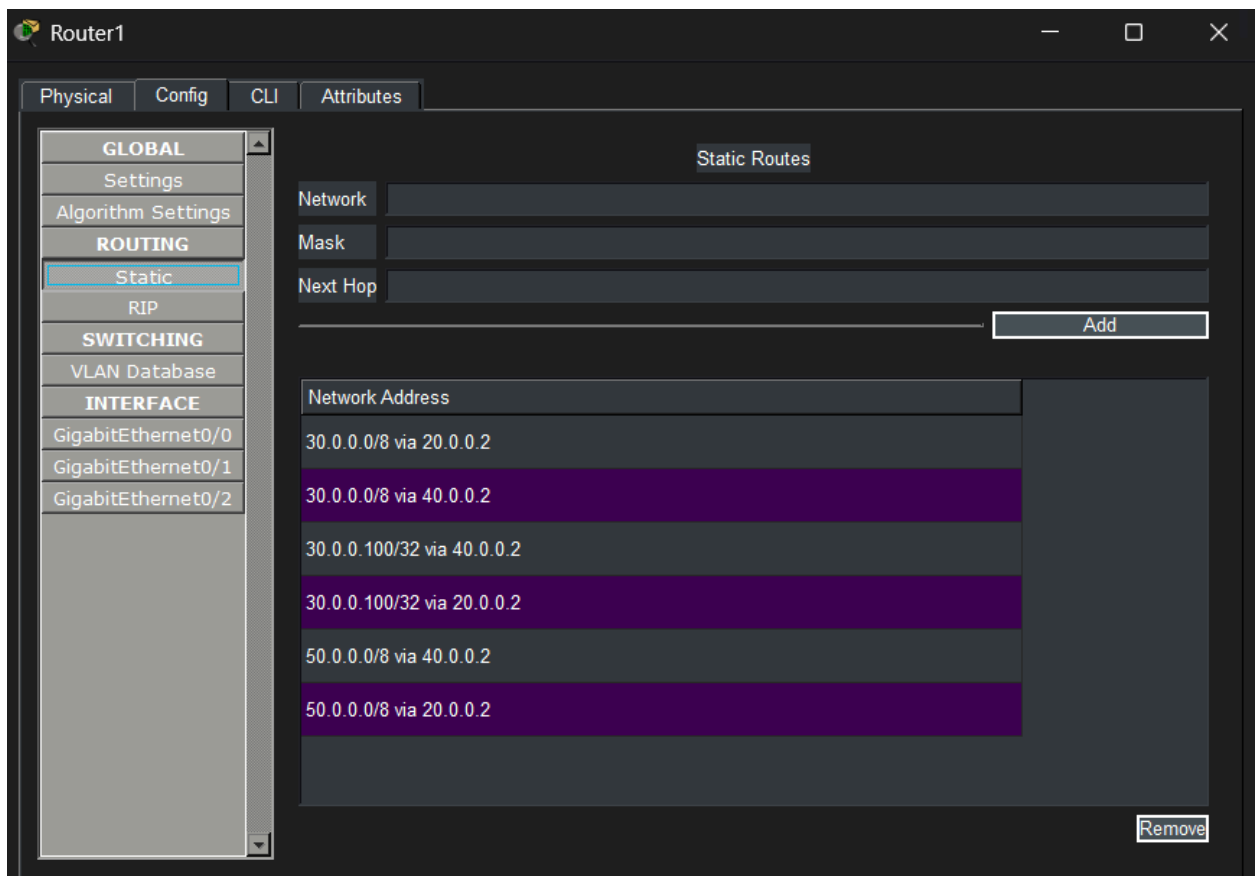
Task 2: Implementing Static Routing in a Three-Router Configuration

1. Construct the topology shown in Figure 6.



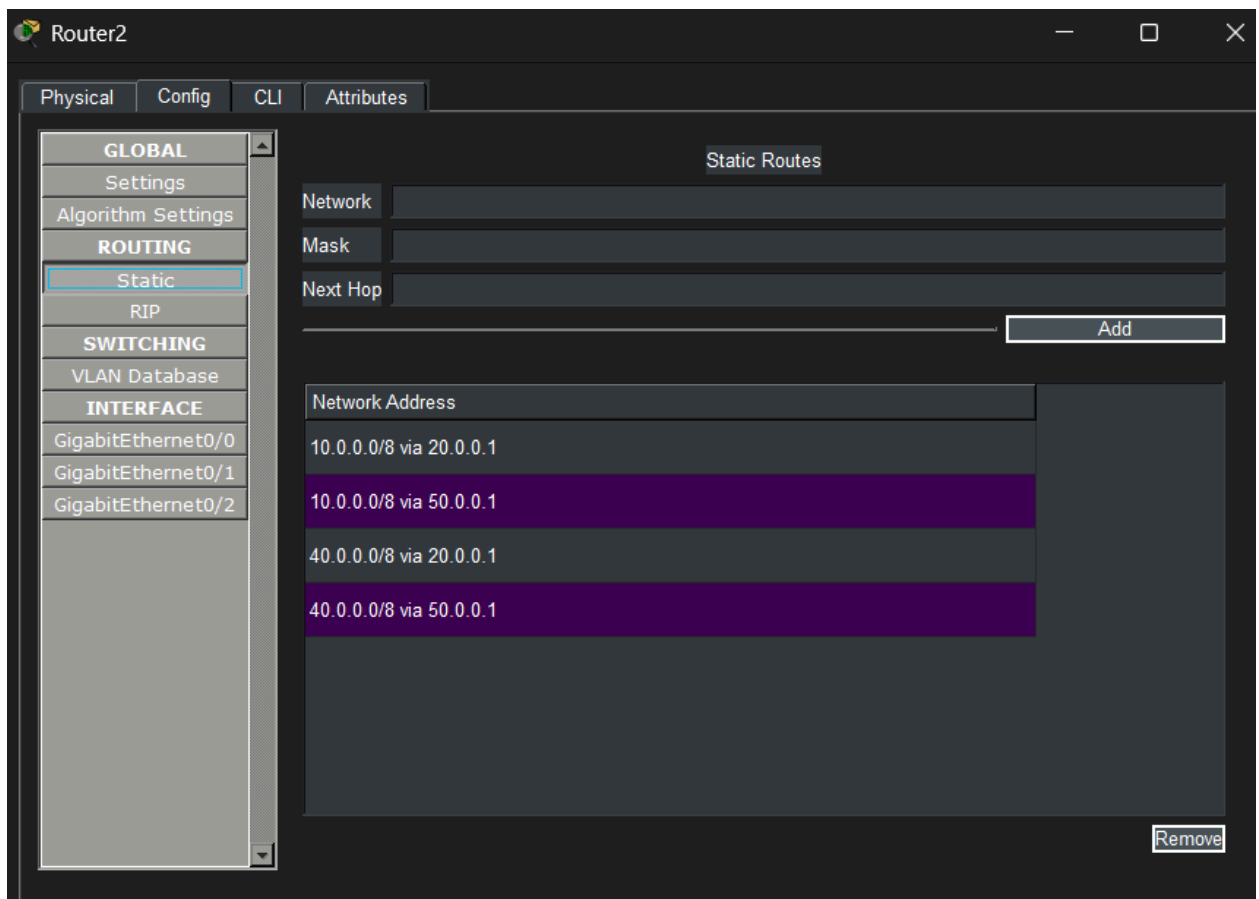
2. Assign IP addresses, subnet masks and default gateway addresses according to Table 2.
3. Configure static routing on router 1,
 - a. Create two routes for network 30.0.0.0/8 and configure the first route (via -Router2) as the main route and the second route (via-Router3) as a backup route.
 - b. Create two routes for the host 30.0.0.100/8 and configure the first route (via -Router3) as the main route and the second route (via-Router2) as a backup route.
 - c. Create two routes for network 50.0.0.0/8 and configure the first route (via -Router3) as the main route and the second route (via-Router2) as a backup route.

```
Router>show ip route static
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.100/32 [1/0] via 40.0.0.2
S       50.0.0.0/8 [1/0] via 40.0.0.2
```



4. Configure static routing on router 2,
 - a. Create two routes for network 10.0.0.0/8 and configure the first route (via -Router1) as the main route and the second route (via-Router3) as a backup route.
 - b. Create two routes for network 40.0.0.0/8 and configure the first route (via -Router1) as the main route and the second route (via-Router3) as a backup route.
 - c. Verify the router adds only main routes to the routing table using the “show ip route static” command.

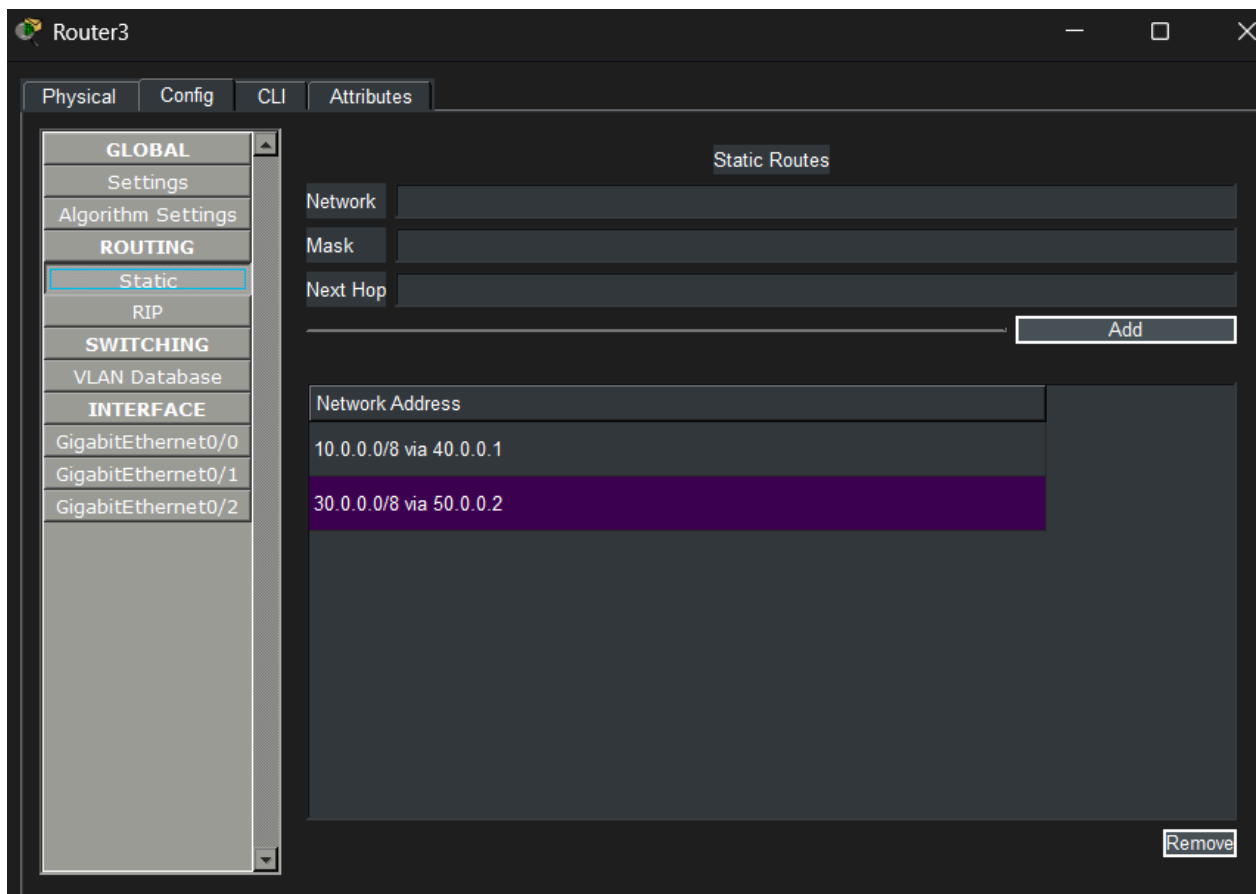
```
Router>show ip route static
S    10.0.0.0/8 [1/0] via 20.0.0.1
S    40.0.0.0/8 [1/0] via 20.0.0.1
```



5. On Router 3, create static routes for network 10.0.0.0/8 and network 30.0.0.0/8 and verify the router adds both routes to the routing table.

```
Router>show ip route static
S    10.0.0.0/8 [1/0] via 40.0.0.1
S    30.0.0.0/8 [1/0] via 50.0.0.2
```

-



6. On Router1, we configured two routes for network 30.0.0.0/8. These routes are via Router2 and via Router3. We set the first route (via-Router2) as the main route and the second route as the backup route. We can verify this configuration in two ways.
 - By sending ping requests to a PC of network 30.0.0.0/8 and tracing the path they take to reach the network 30.0.0.0/8. For this, you can use '**tracert**' command on a PC of network 10.0.0.0/8. The '**tracert**' command sends ping requests to the destination host and tracks the path they take to reach the destination.
 - By listing the routing table entries on Router0. Since a router uses the routing table to forward data packets, you can check the routing table to figure out the route the router uses to forward data packets for each destination.

Verify static routing by using the **tracert** command on PC1 to ping Server S0.



```
Tracing route to 30.0.0.100 over a maximum of 30 hops:
```

1	0 ms	0 ms	0 ms	10.0.0.1
2	0 ms	0 ms	0 ms	40.0.0.2
3	0 ms	0 ms	0 ms	20.0.0.2
4	0 ms	0 ms	0 ms	30.0.0.100

```
Trace complete.
```



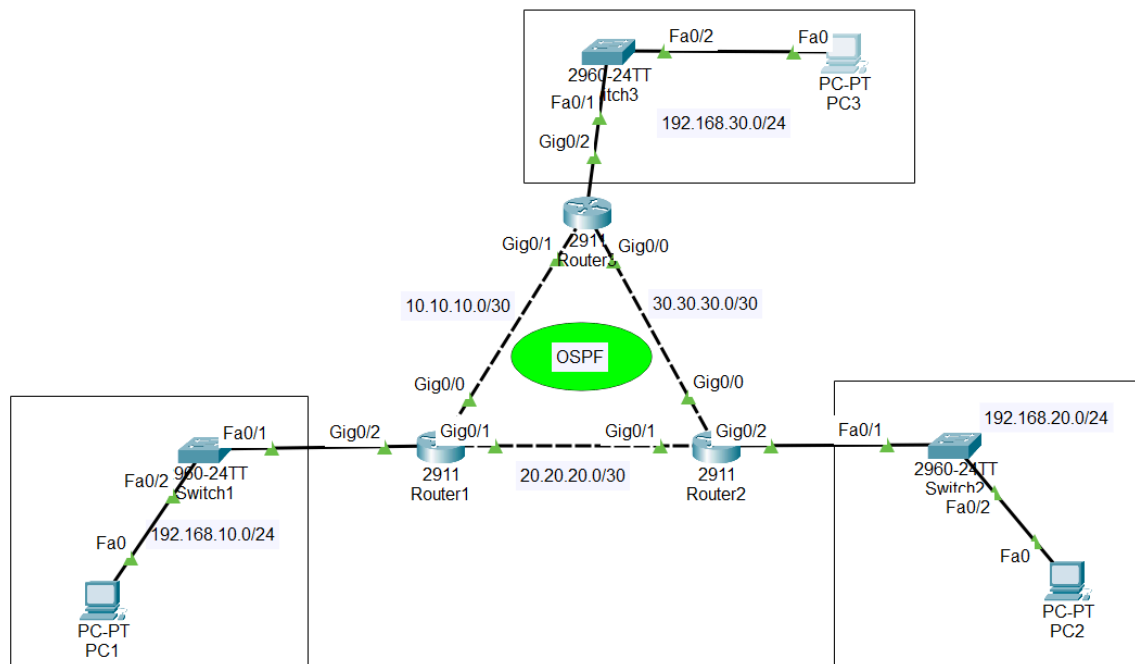
7. Use the **tracert** command on PC1 to trace the route taken by a packet sent to PC3

```
Tracing route to 30.0.0.1 over a maximum of 30 hops:

 1  0 ms      0 ms      0 ms      10.0.0.1
 2  0 ms      3 ms      0 ms      30.0.0.1

Trace complete.
```

Topology



.Figure 7

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/0	10.10.10.1	255.255.255.252	N/A	N/A
	G0/1	20.20.20.1	255.255.255.252	N/A	N/A



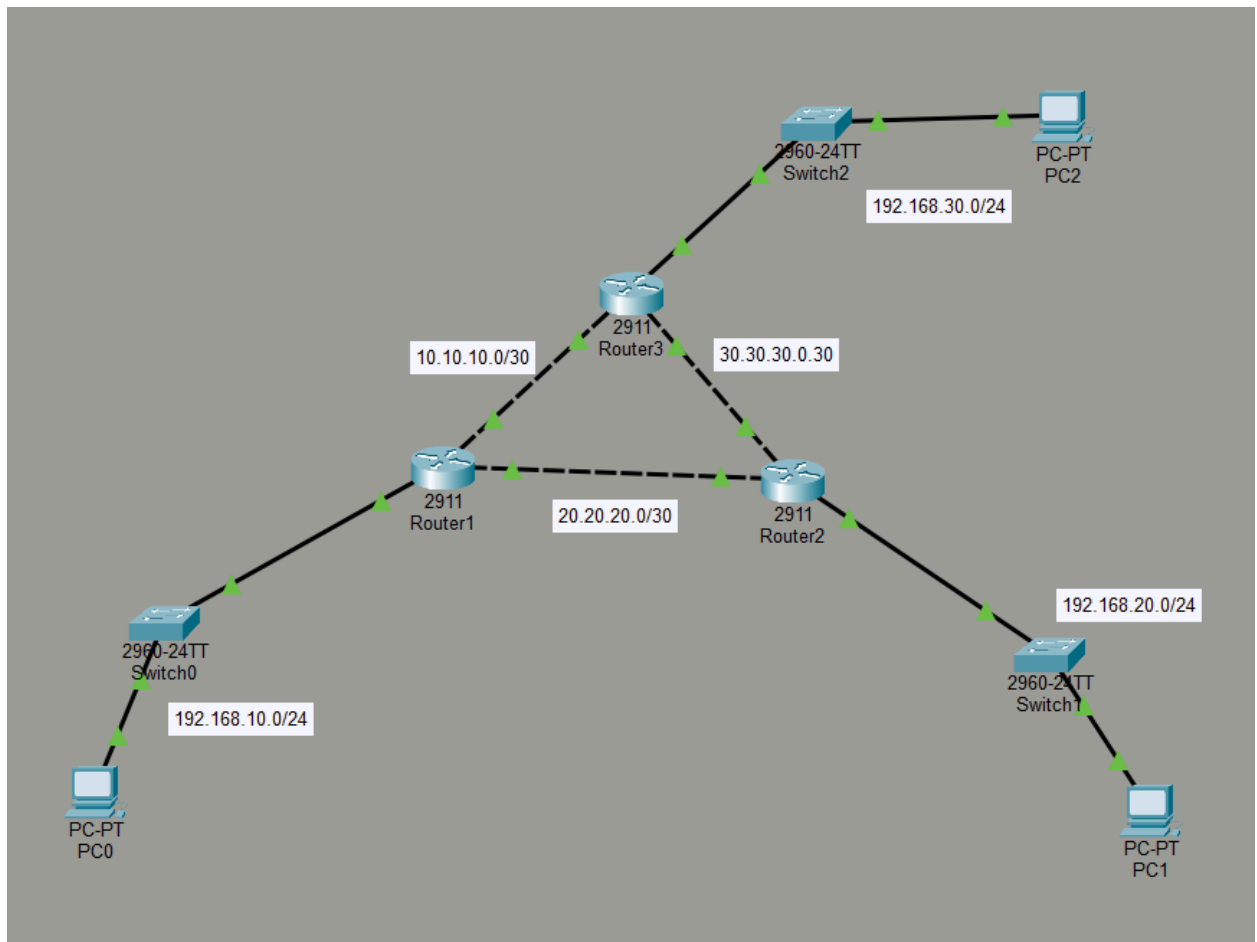
	G0/2	192.168.10.1	255.255.255.0	N/A	S1 F0/1
R2	G0/0	30.30.30.1	255.255.255.252	N/A	N/A
	G0/1	20.20.20.2	255.255.255.252	N/A	N/A
	G0/2	192.168.20.1	255.255.255.0	N/A	S2 F0/1
R3	G0/0	30.30.30.2	255.255.255.252	N/A	N/A
	G0/1	10.10.10.2	255.255.255.252	N/A	N/A
	G0/2	192.168.30.1	255.255.255.0	N/A	S3 F0/1
PC-1	NIC	192.168.10.10	255.255.255.0	192.168.10.1	S1 F0/2
PC-2	NIC	192.168.20.10	255.255.255.0	192.168.20.1	S2 F0/2
PC-3	NIC	192.168.30.10	255.255.255.0	192.168.30.1	S3 F0/2

Task 3 - IP Addressing Table

Task 3: Configuring OSPF for Dynamic Routing

Open Shortest Path First (OSPF) is a link-state routing protocol used to manage IP routing within an autonomous system. It operates by having routers exchange information about the state of their network links to build a comprehensive map of the network topology.

1. Construct the topology shown in Fig 7.



2. Assign addresses according to Table 3. Ensure each device is configured correctly with the assigned IP address and subnet mask.
3. Enable OSPF routing on router 1 in global configuration mode by using the command below.

```
Router(config)# router ospf [process_ID]
```

The process_ID is a unique identifier for the OSPF process on this router, and it can be any number from 1 to 65,535. Note that the process_ID does not need to match across different routers, as it is locally significant to each router.

4. Assign an area to the interface

```
Router(config-router)# network IP_network_# [wildcard_mask] area [area number]
```




This command enables OSPF on the interface if the IP address matches the specified network. The wildcard_mask is used to identify which parts of the IP address to match, where a 0 indicates a matching bit and a 1 indicates an ignoring bit. For example, a wildcard mask of 0.255.255.255 will match only the first byte of the network address.

The area_number specifies the OSPF area that the interface will be part of. It can range from 0 to 4,294,967,295. OSPF areas are used to manage and limit the routing information shared between routers. Detailed routing information is shared only within the same OSPF area, while summarized information is shared between different OSPF areas. Note that Area 0 is the backbone area, and all other OSPF areas must connect to it.

5. Next, configure OSPF on router 1 to advertise the connected networks. The configuration for Router 1 is shown below.

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.10.0 0.255.255.255 area 0
Router(config-router)#
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 20.20.20.0 0.0.0.3 area 0
Router(config-router)#

Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.10.0 0.255.255.255 area 0
Router(config-router)#
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 20.20.20.0 0.0.0.3 area 0
Router(config-router)#
Router(config-router)#
```

6. Repeat steps 3-5 for Router 2 and 3.
Router 2:



```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.20.0 0.255.255.255 area 0
Router(config-router)#network 20.20.20.0 0.0.0.3 area 0
Router(config-router)#network 20.0.30.0 .
02:32:47: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on GigabitEthernet0/1 from LOADING to FULL,
Loading Done
% Incomplete command.
Router(config-router)#network 20.0.30.0 0.0.0.3 area 0
Router(config-router)#30.30.30.0 0.0.0.3 area 0
^
% Invalid input detected at '^' marker.

Router(config-router)#network 30.30.30.0 0.0.0.3 area 0
Router(config-router)#exit
Router#
```

Router 3:

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#192.168.30.0 0.255.255.255 area 0
^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.30.0 0.255.255.255 area 0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 30.30.30.0 0.0.0.3 area 0
Router(config-router)#
02:36:33: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on GigabitEthernet0/1 from LOADING to FULL,
Loading Done

Router(config-router)#
02:36:41: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.20.1 on GigabitEthernet0/0 from LOADING to FULL,
Loading Done
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

7. After configuring OSPF on all routers, verify the configuration by checking the OSPF routing table using the **show ip route ospf** command and neighbor status using **show ip ospf neighbor**



```
Router>show ip route ospf
      30.0.0.0/30 is subnetted, 1 subnets
O       30.30.30.0 [110/2] via 10.10.10.2, 00:00:52, GigabitEthernet0/0
          [110/2] via 20.20.20.2, 00:00:52, GigabitEthernet0/1
O      192.168.20.0 [110/2] via 20.20.20.2, 00:04:53, GigabitEthernet0/1
O      192.168.30.0 [110/2] via 10.10.10.2, 00:01:02, GigabitEthernet0/0

Router>show ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.30.1	1	FULL/BDR	00:00:34	10.10.10.2	GigabitEthernet0/0
192.168.20.1	1	FULL/BDR	00:00:34	20.20.20.2	GigabitEthernet0/1

```
Router>
```

8. Ping from one end device to another across the network to ensure OSPF is routing traffic correctly.

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

What if now, an unauthorized router starts flooding your network with false advertisements, threatening its stability? In our next lab, we'll tackle this issue by setting up OSPF authentication to secure our network.



Assessment Rubric
Lab 02
Introduction to Packet Tracer and IP Routing

Name:	Student ID:
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Points Distribution

Task No.	LR 2 Simulation	LR9 Report
Task 1	20	
Task 2	30	
Task 3	30	
Total	/80	/20
CLO Mapped	CLO 1	CLO1

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/20	CLO 1

CLO	Total Points	Points Obtained
1	100	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

**Lab Evaluation Assessment Rubric**

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR2	Program/Code / Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.